

Smart Metro Ticket Generation System

Phase 1.1 | Problem Statement

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	2 Marks

This document defines the **Problem Statement** for Phase 1.1 of the Smart Metro Ticket Generation System — a ServiceNow-based application for metro ticket booking, passenger management, fare calculation and ticket generation. It converts the raw ideas gathered during brainstorming into a single, precise statement of the problem the team is committing to solve.

Purpose of This Phase

- Translate scattered pain points and observations into one clear, agreed problem
- Give the whole team a shared, unambiguous target before any solution design begins
- Make it possible to later check every proposed feature against a single yardstick — does it actually address this problem?
- Provide evaluators with a crisp statement of intent for this phase of the ideathon

What Makes a Strong Problem Statement

Quality	What It Means in Practice
Specific	Names the exact users and exact moment in the journey affected
Evidence-based	Backed by observations, complaint logs, or direct experience, not assumption
Solution-free	Describes the problem only — it does not smuggle in a feature or fix
Measurable	Impact can be described in a way that shows whether it later improved

Inputs Carried Forward From Brainstorming

Phase 1.1 builds directly on the idea pool and themes captured in Phase 1.2 brainstorming — commuter queue delays, manual fare errors, scattered passenger records, paper-ticket dependency, and limited real-time visibility for staff. These recurring threads are converged below into a single problem statement.

02 The Problem Statement

Commuters using the metro today face a fragmented, manual ticketing experience. Long queues at counters, error-prone manual fare calculation, absence of a centralized passenger record, and lack of automation increase wait times and operational overhead for metro staff. There is no unified platform that ties booking, passenger data, fare logic, and ticket issuance together in one automated, auditable workflow.

Core Statement

How might metro operations replace slow, manual, error-prone ticketing with a single automated ServiceNow-based system that handles booking, passenger management, fare calculation and ticket generation end-to-end — reducing wait time, human error, and administrative load?

Observed Pain Points

Pain Point	Impact	Affected Stakeholder
Long queues at manual counters	Increased boarding time, commuter frustration	Passenger
Manual fare calculation errors	Revenue leakage, disputes	Operator / Passenger
No centralized passenger records	Difficult to track passes, history, refunds	Operator
Paper ticket dependency	Higher printing cost, environmental impact	Operator
No real-time visibility for staff	Slow issue resolution, poor reporting	Admin / Staff
Inconsistent refund handling	Commuter distrust, repeated support requests	Passenger / Support

Stakeholder Impact Summary

Stakeholder	How the Problem Affects Them
Daily Commuter	Loses time in queues, risks missing trains, has no visibility into fare logic
Counter / Gate Staff	Repeats manual tasks, handles avoidable disputes over fare or refunds
Metro Administration	Lacks real-time data to plan staffing, capacity, or spot revenue leakage

03 Root Cause & Path Forward

Before moving to solution design, the team traced the problem back to its underlying causes using a simple '5 Whys' pass — this keeps the eventual solution focused on the actual root cause rather than a visible symptom.

Question	Answer
Why do commuters face long queues?	Ticketing still relies on manual counters for most passengers
Why does it rely on manual counters?	No self-service digital booking option is available end-to-end
Why is there no digital option?	Booking, fare and ticketing were never unified onto one platform
Why were they never unified?	Each function was historically built or operated as a separate manual step
Why does that matter (root cause)?	Without one automated system, every stage stays manual, slow and error-prone

Supporting Evidence

- Commuter complaint logs repeatedly cite queue time and fare disputes as top frustrations
- Staff shadowing at counters showed the same fare lookup steps repeated for every passenger
- No existing system currently links passenger identity, fare rules and ticket issuance together

Consequences If Left Unresolved

- Continued commuter dissatisfaction and slower boarding during peak hours
- Ongoing revenue leakage from manual fare miscalculation
- Rising operational cost from paper tickets and manual record-keeping
- Missed opportunity to use passenger data for better service planning

Next Steps

- Build the commuter empathy map to validate this statement against real user experience
- Carry the root cause forward into the Recommended Solution Design phase
- Keep this statement visible throughout the ideathon as the single source of truth

Closing Note

This problem statement is deliberately solution-free. It exists to keep every later decision — from empathy mapping to feature prioritization — anchored to one real, evidence-backed problem rather than a list of desired features.